# ISC 2, Checkpoint 2

## YOLO

### References

- https://www.v7labs.com/blog/yolo-object-detection
- https://www.datacamp.com/blog/yolo-object-detection-explained

### Conversation with Gemini

https://notebooklm.google.com/notebook/babc6073-543f-44f5-bc59-249a43e622b8

### Problem specification:

#### Input

- Image: $X \in \mathbb{R}^{C \times H \times W}$
- List of possible classes $\mathcal{C} = \{c_1, c_2, ..., c_m\}$

#### Output

- Bounding boxes

#### 1. Grid division

- Divide the image into a grid $S \times S$. Box at row $i$ column $j$ denoted as $\text{Cell}^{(ij)}$.

#### 2. Bounding box regression

- For each grid cell, calculate the bounding box within the cell

$$\text{Cell}^{(ij)} \xrightarrow{f_1} \left[\boldsymbol{B}^{(ij,k)} \mid \boldsymbol{C}^{(ij,k)}\right] = \begin{bmatrix} p^{(k)} & b_x^{(k)} & b_y^{(k)} & b_w^{(k)} & b_h^{(k)} & \mid & c_1^{(k)} & c_2^{(k)} & \dots & c_m^{(k)} \end{bmatrix}_{B \times (5+m)}$$

| **Variable** | *Description* | *Range* |
|---|---|---|
| $p$ | probability/confidence of $\text{Cell}^{(ij)}$ containing an object | $(0, 1)$ |
| $b_{[.]}$ | relative centre location and dimensions of the bounding box<br>• actual $x = \sigma(b_x) + g_x$<br>• actual $y = \sigma(b_y) + g_y$<br>• actual $w = g_w e^{b_w}$<br>• actual $h = g_h e^{b_h}$ | $(0, 1)$ |
| $c_j$ | $\mathbb{P}[\text{class } c_j \mid \text{object}]$ | $(0, 1)$ |

#### 3. Non-Maximum Supression

- Sort bounding boxes by confidence, select top $k$
- Select based on an IoU threshold

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### A. Design decisions

- Coupled vs decoupled head.

## Ultralytics's YOLO `Results` object

- https://docs.ultralytics.com/reference/__init__/
- https://docs.ultralytics.com/modes/
- `boxes: Boxes`
- `masks: Masks`